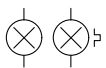
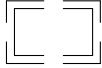
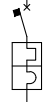
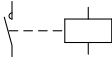
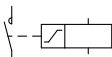
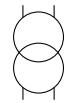
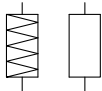
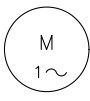
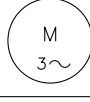


1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																					
SAFETY CRITICAL ELEMENT LIST																					
IFI	2	11/04/2025	AS-BUILT				ADF	GDP	IC												
IFI	1	31/03/2025	GENEERAL REVIEW				ADF	GDP	IC												
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC												
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.												
	Rev. Index	Date	Description				Prepared	Checked	Approved												
Client							Project Title														
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT														
Contractor							Provision of a m-LNG plant for production for ManaEnergy														
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																					
Engineering																					
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																					
Facility Name			Location				Scale		Sheet of Sheets												
TENDRARA			MOROCCO				A4		1 of 20												
Document Title					Doc. Code																
P-PN-S221B.1					ELD01-ING-FR01																
LNG LOADING BAY S-221B					Vendor Doc. code																
					IE24090_SE03_P-PN-S221B.1																
			Project : P–PN–S221B.1				Customer : ManaEnergy Managem Group				Object :										
General Contractor :		Manufacture : esa esaelettromeccanica.com		LNG LOADING BAY S–221B				Contract : IE24090				Rev. 2		Data : 11/04/2025				Date : 26/03/2024			
								S/N : IE24090_SE03				Rev. 1		Data : 31/03/2025				Sheet : 1		of : 20	

CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

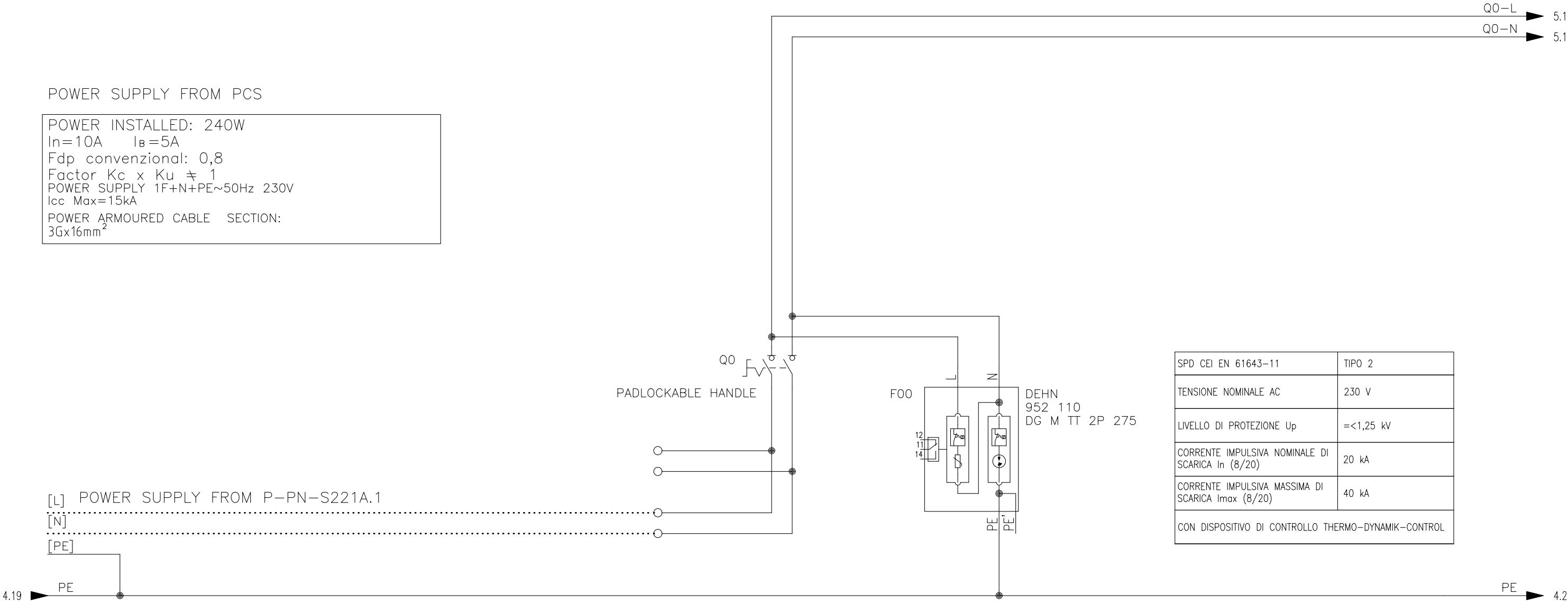
CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
INTRINSIC SAFE SIGNAL	BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

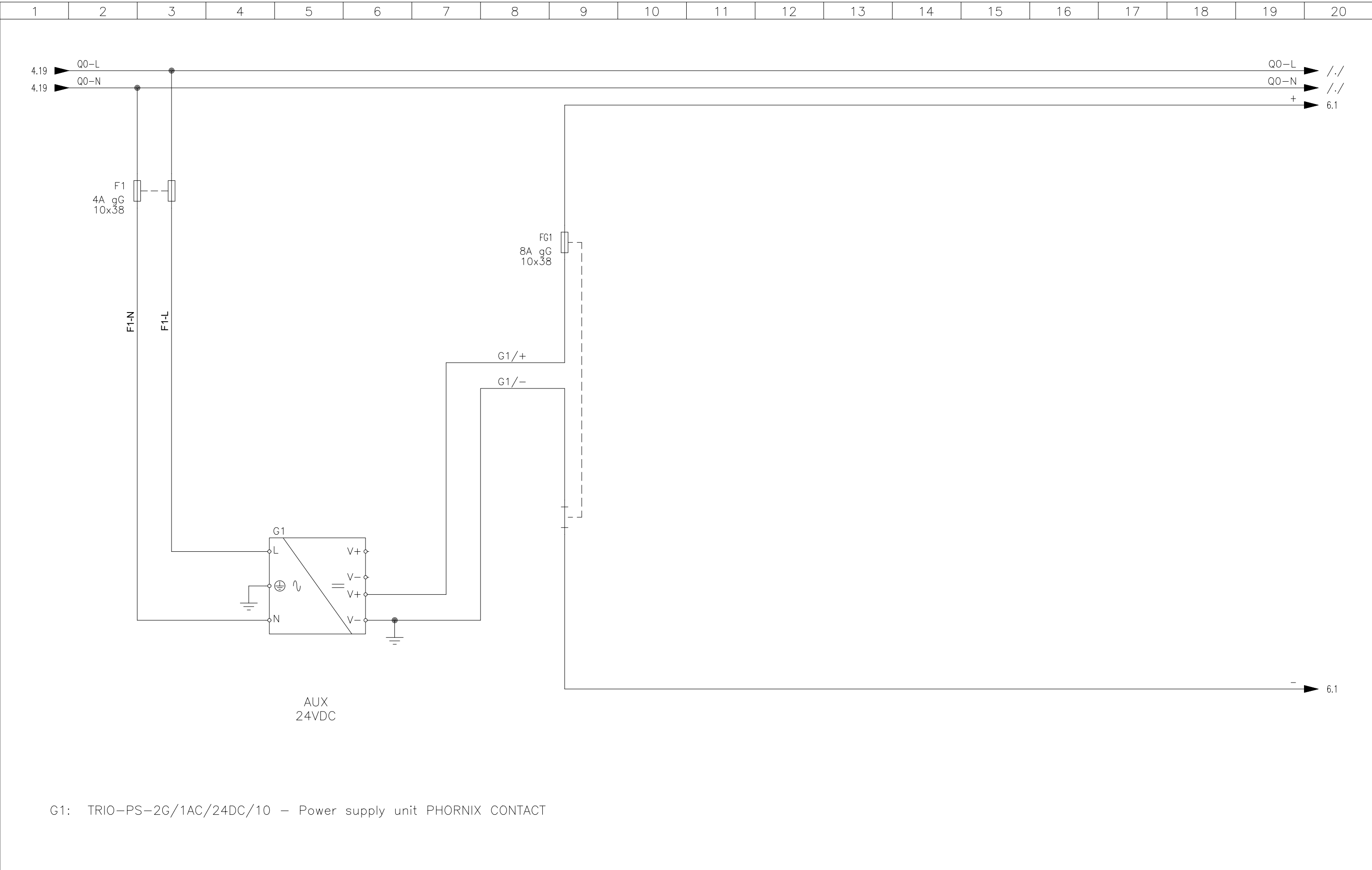
	POWER MAIN DISCONNECT SWITCH																	FEEDER ID	
	1F+N+PE~50Hz 230V																	RATED POWER Pn-RATED VOLTAGE Vn	
																		RATED CURRENT	

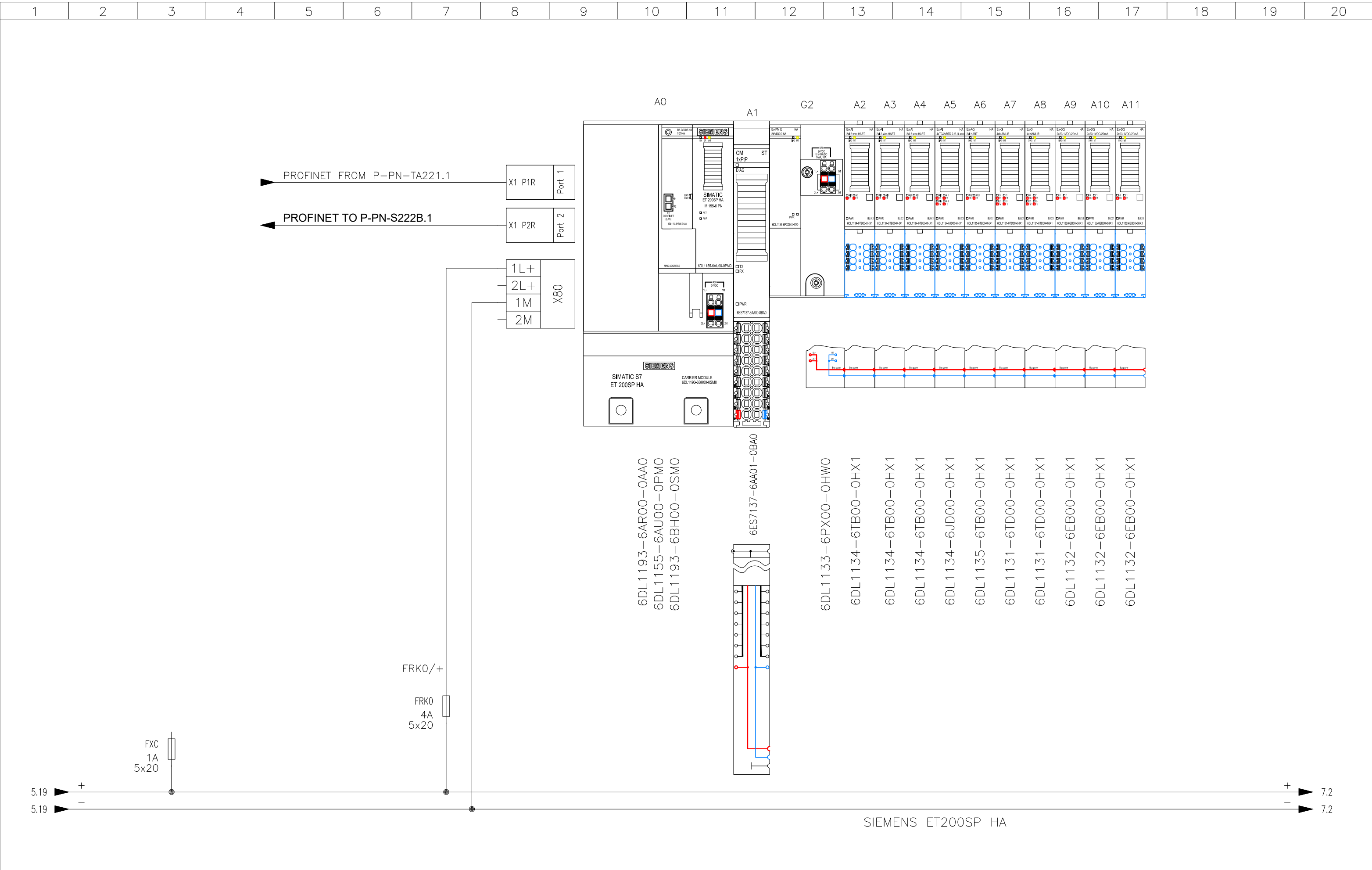
POWER SUPPLY FROM PCS

POWER INSTALLED: 240W
In=10A IB=5A
Fdp convenzional: 0,8
Factor Kc x Ku ≈ 1
POWER SUPPLY 1F+N+PE~50Hz 230V
Icc Max=15kA
POWER ARMoured CABLE SECTION:
3Gx16mm²



CONDUCTOR PROTECTION	TECHNOR PSRC-220										TYPE	CONDUCTOR PROTECTION
	20A										RATED BREAKING CAPACITY Icu	
											RATED or THERMAL CURRENT In or Ith	
											MAGNETIC RELEASE CHARACTERISTICS Im	
											RATED R.C. SENSITIVITY Idn	
											WIRE SECTION	
											CABLE TYPE	
											CABLE CARRYING LOAD Iz	
											CABLE LENGHT	
											LINE ID	
WIRE ID ON TERMINAL												





6.19 +
6.19 -

8.1 +
8.1 -

F2
1A
5x20

F2/+

M

T/RA- 8.12
T/RB+ 8.12

11 12 13 14 15 16

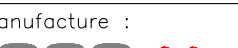
L+ M

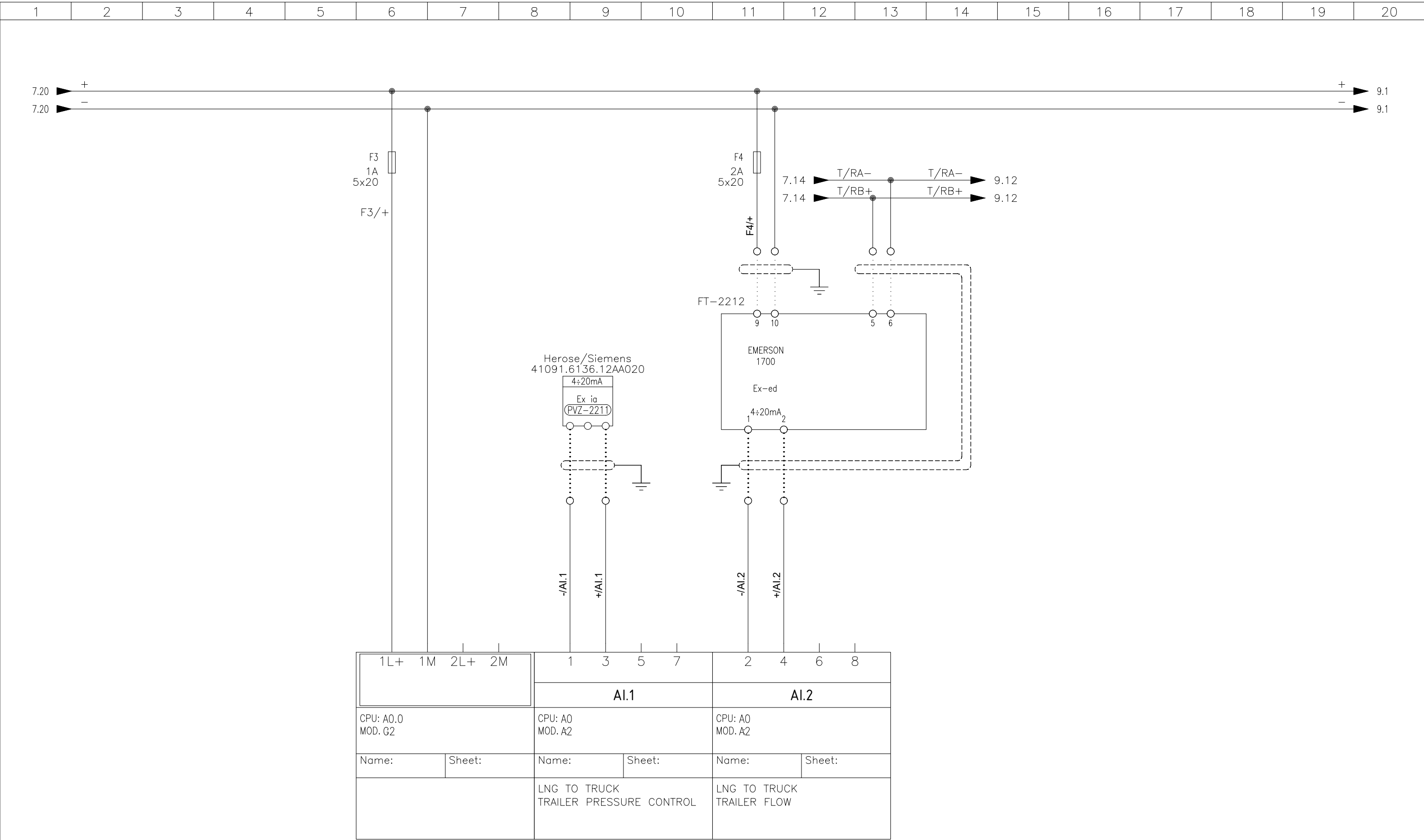
CPU: A0
MOD. A1

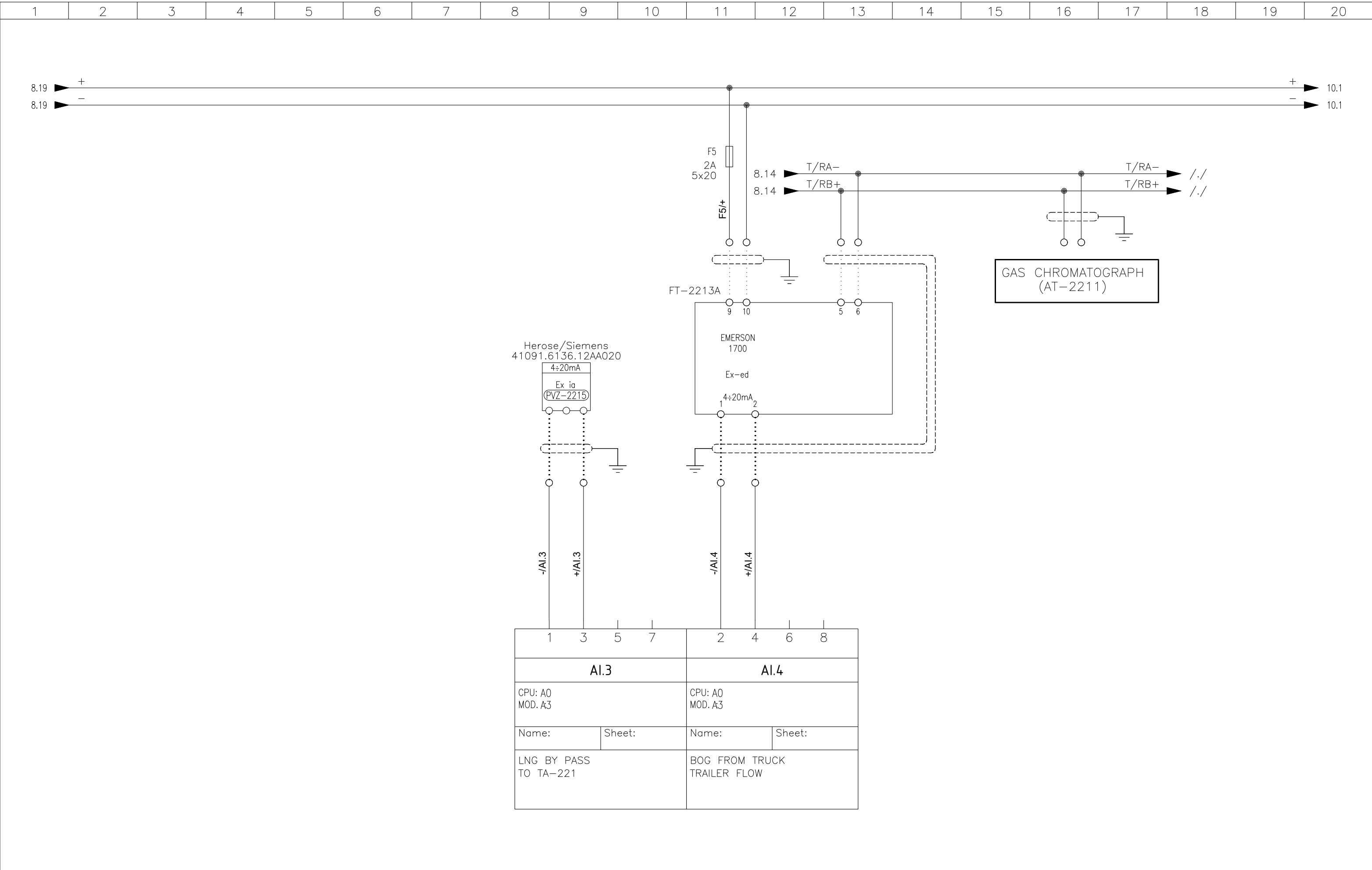
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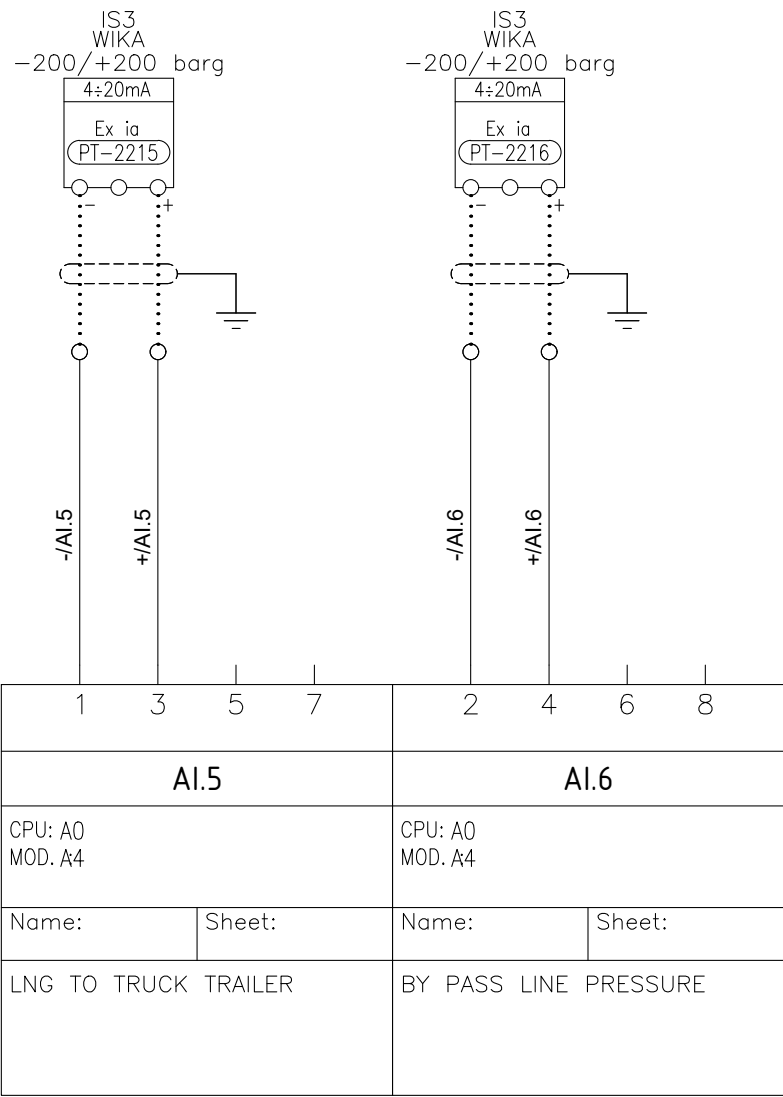
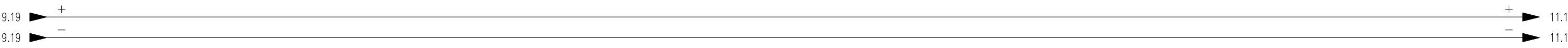
RS-485
TO FLOW TRANSITTER

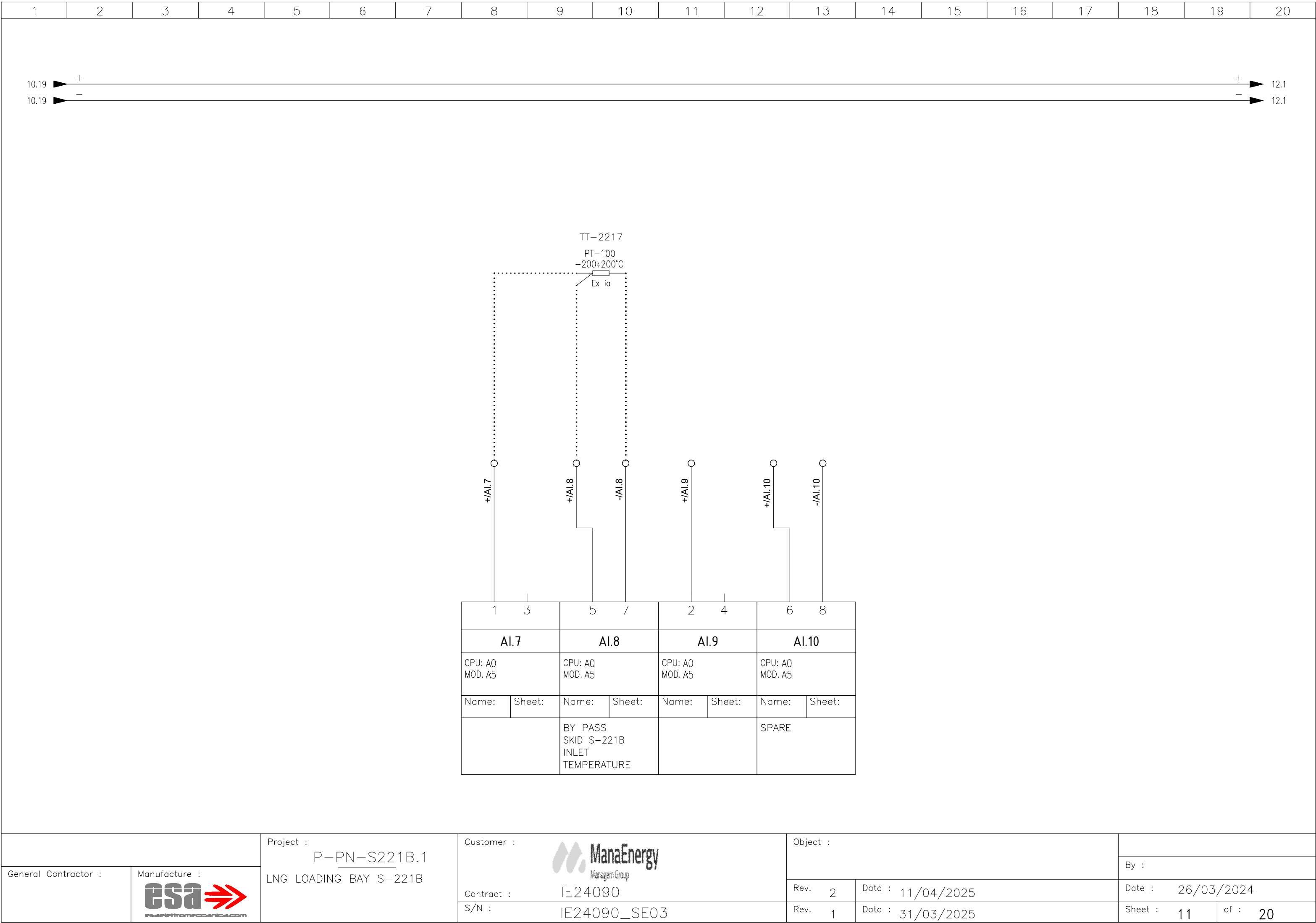
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General Contractor : Manufacture : 		Project : P-PN-S221B.1 LNG LOADING BAY S-221B	Customer :  Contract : IE24090 S/N : IE24090_SE03	Object :		By :	
				Rev. 2	Data : 11/04/2025	Date : 26/03/2024	
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General Contractor :

Manufacture :


Project :

P-PN-S221B.1

LNG LOADING BAY S-221B

Customer :



Contract : IE24090

S/N : IE24090_SE03

Object :

Rev. 2

Data : 11/04/2025

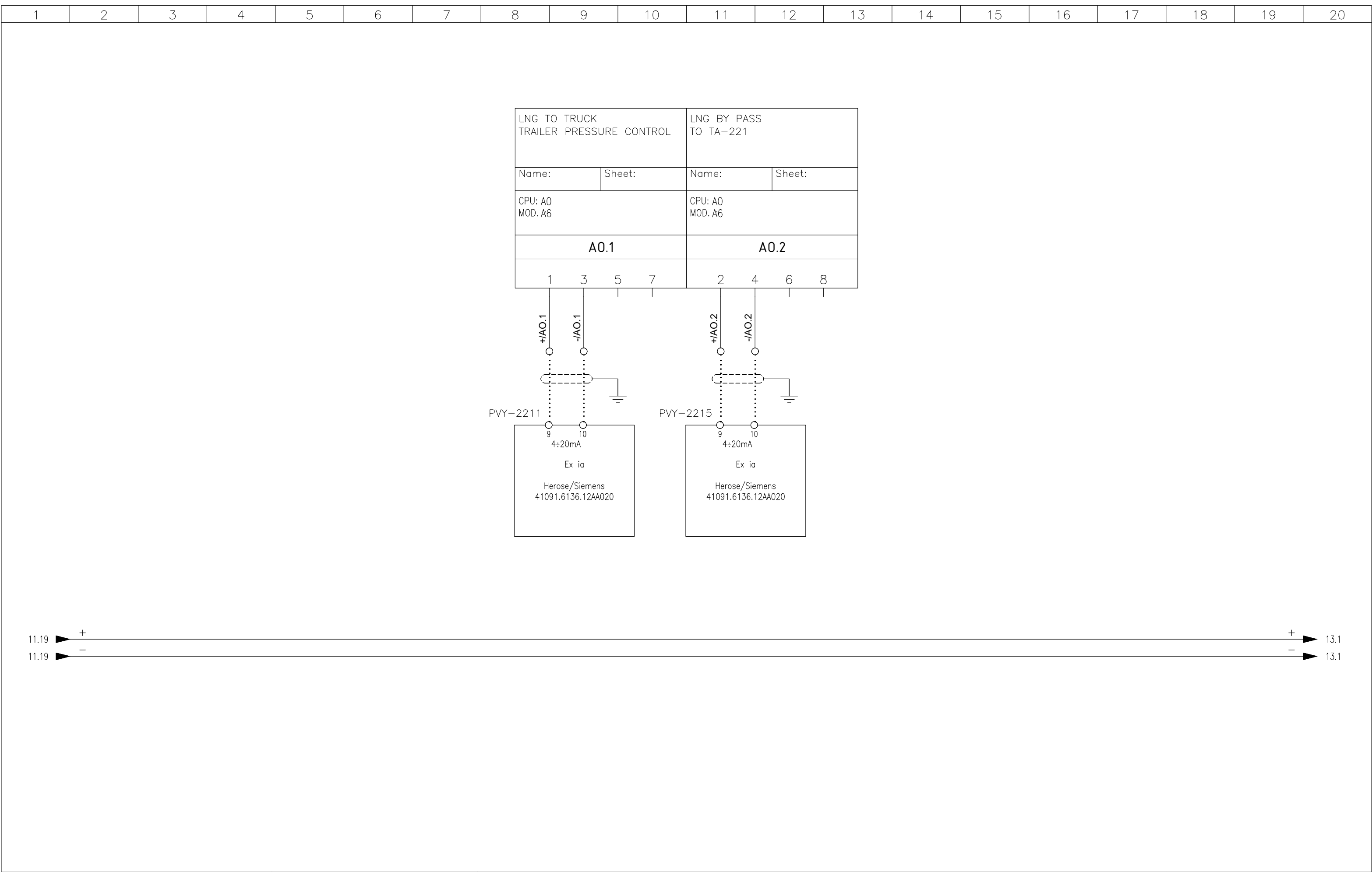
Rev. 1

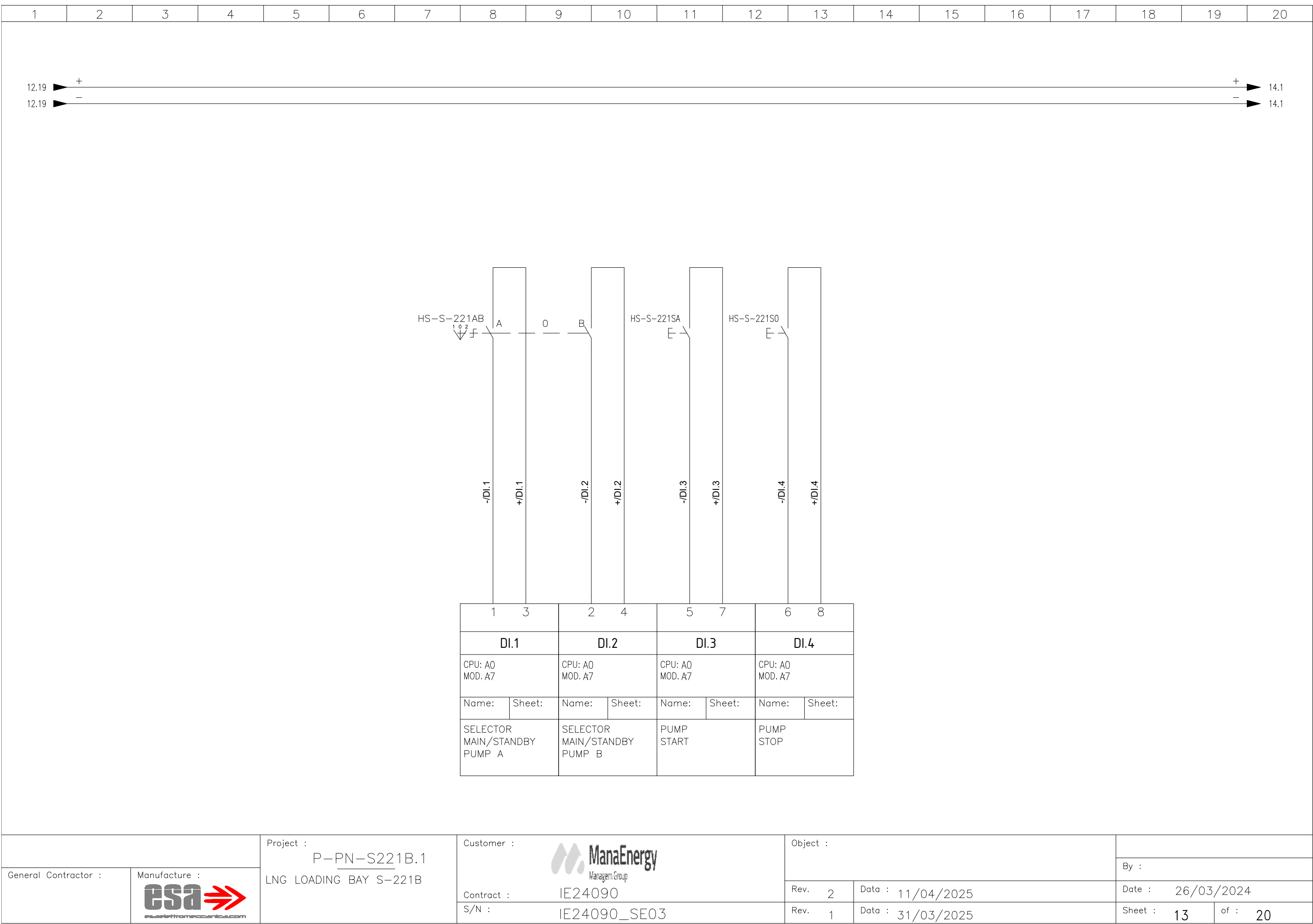
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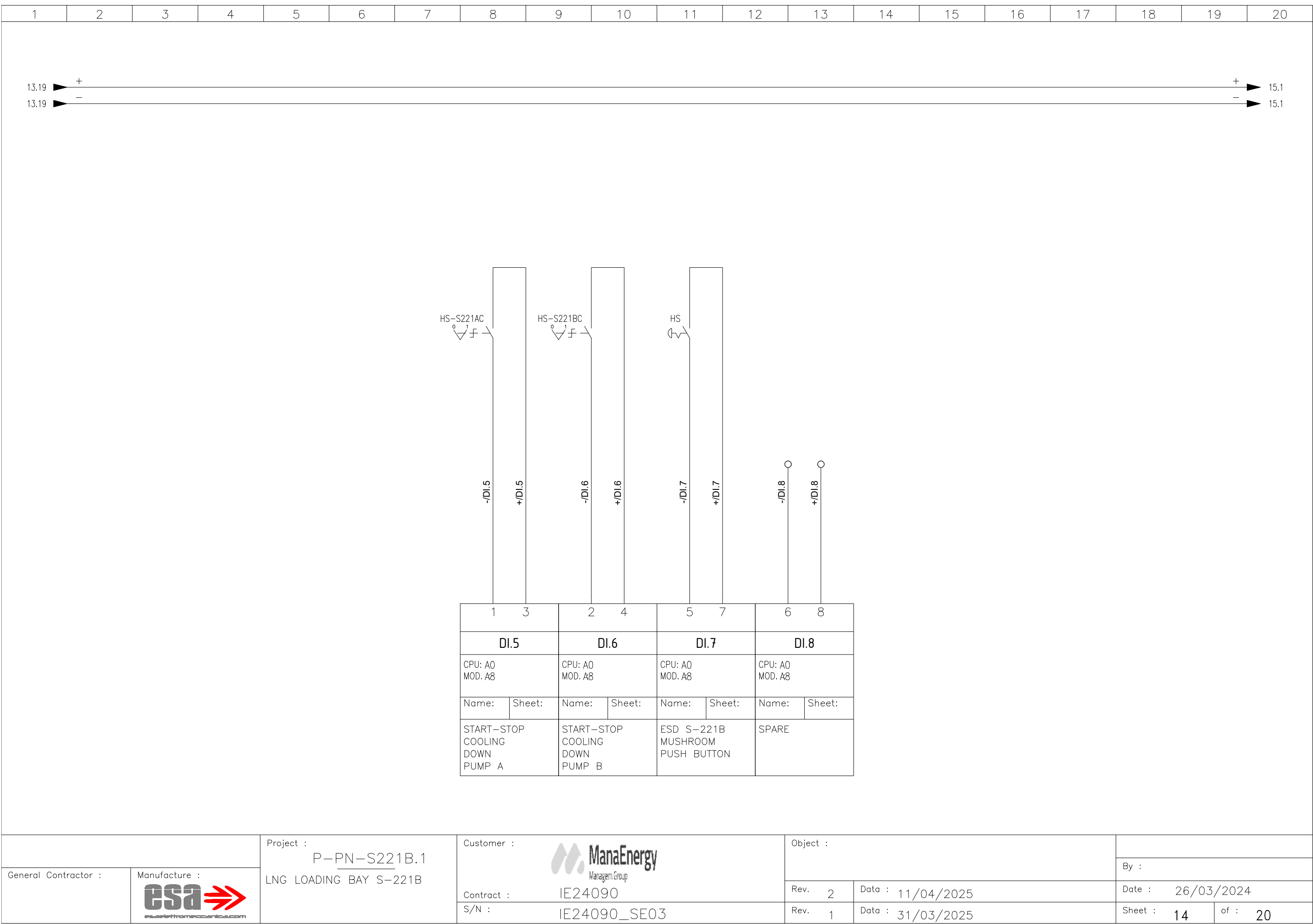
By :

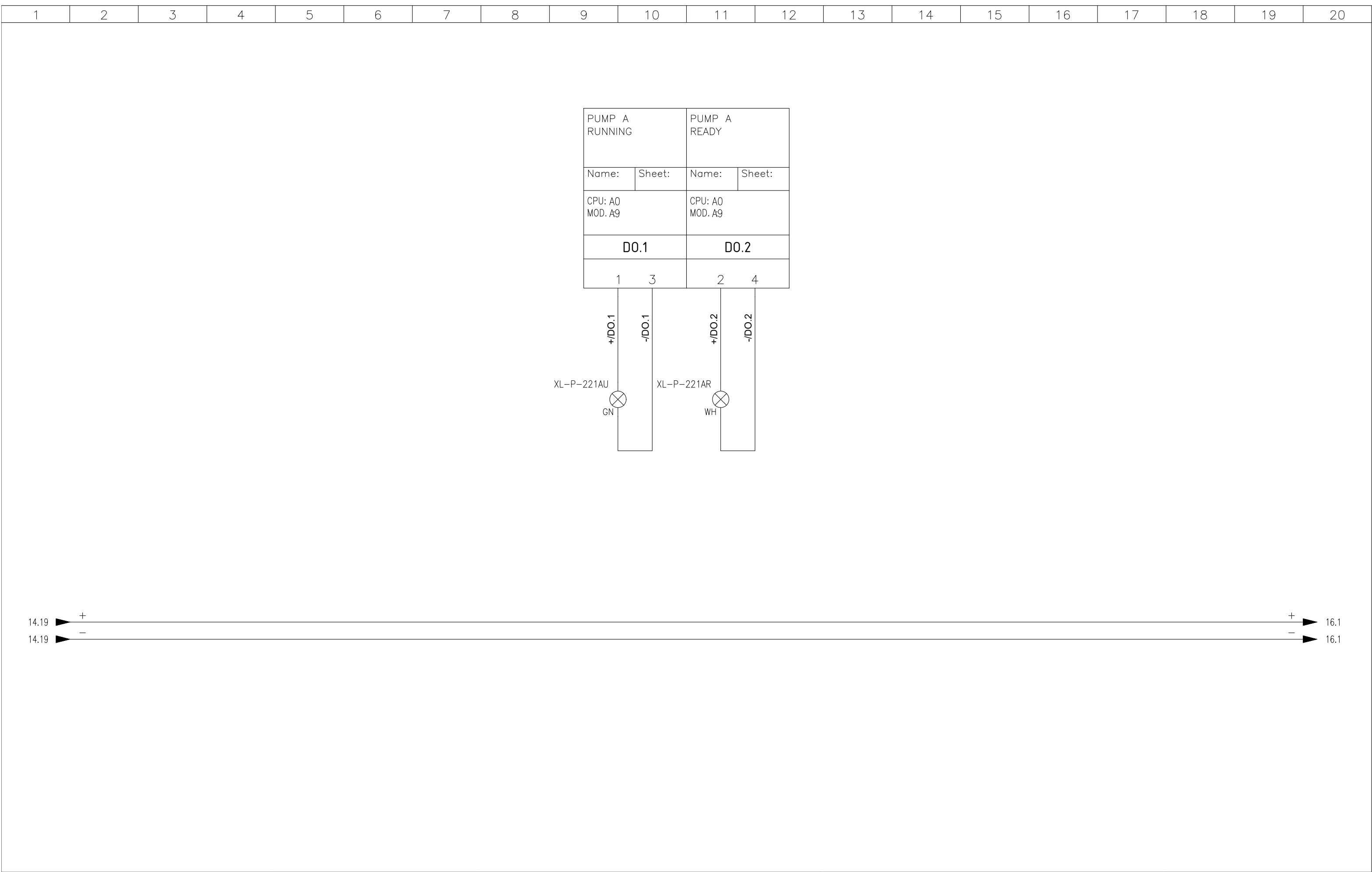
Date : 26/03/2024

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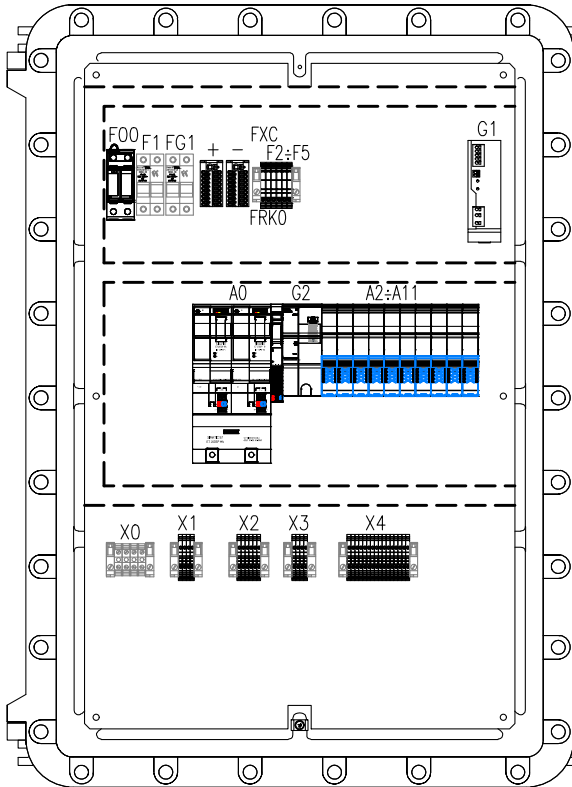
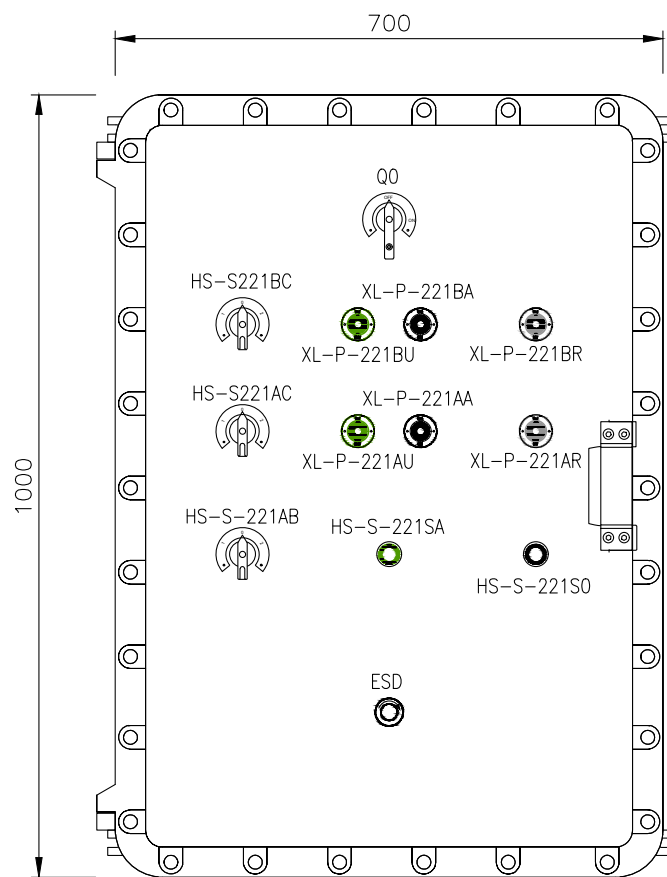






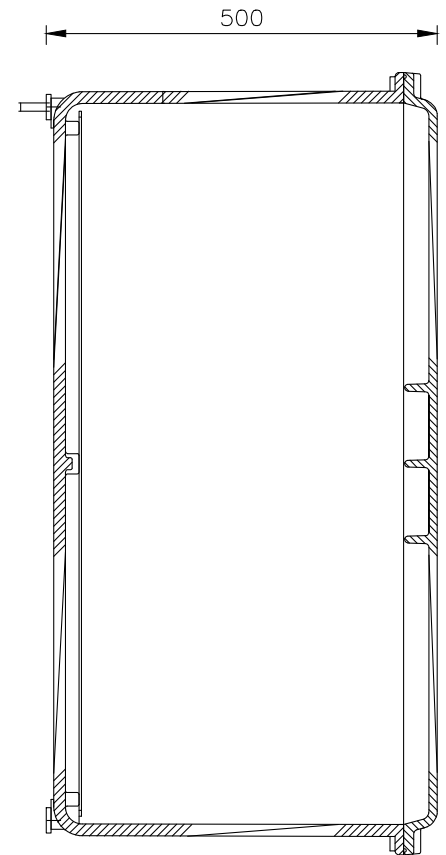
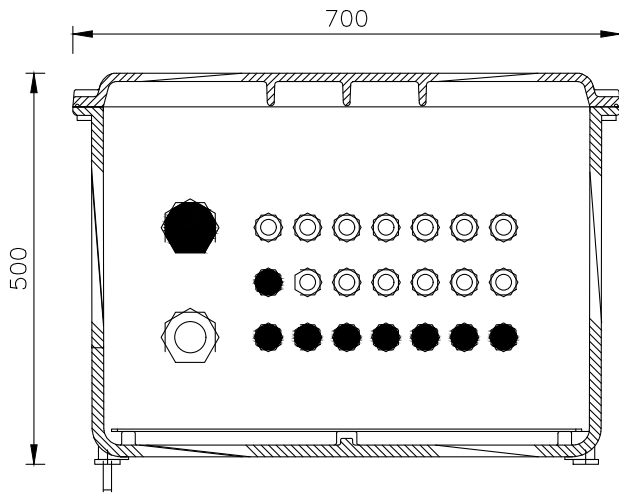
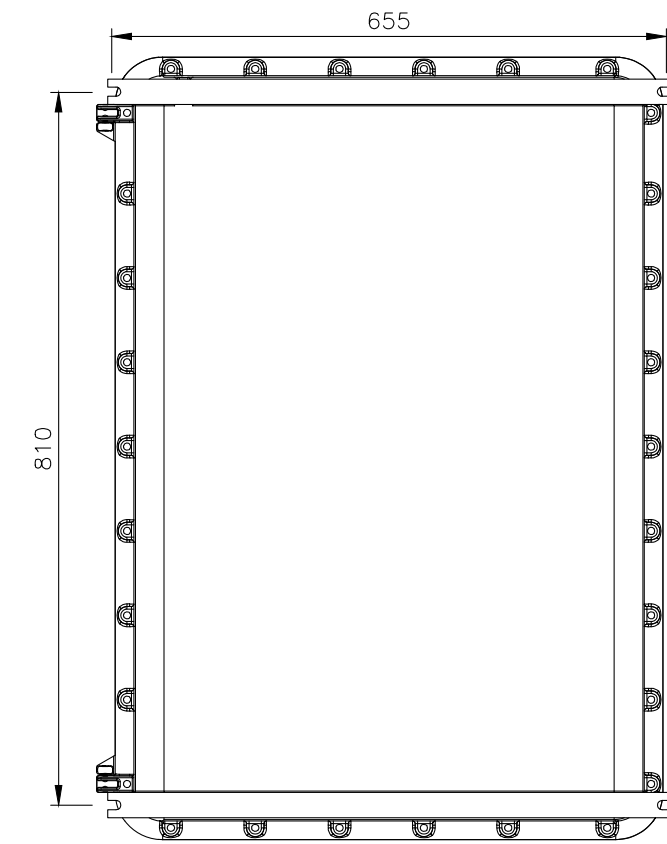






NOTE

- Type : EJB-7
- Ex-d [Ia] IIB T6 IP65 ATEX -20/+50°C
- Installation area is zone 2 IIA T1
- Weight 350kg
- Power dissipation 81W



1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20			
INITIALS	SUPPLIER PN		DESCRIPTION															SUPPLIER				
–	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block															PHOENIX CONTACT				
+	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block															PHOENIX CONTACT				
A0	6DL1193–6MD00–0AA0		”GUIDA PROFILATA 1500MM (59”)”															SIEMENS				
A0	6DL1155–6AU00–0PM0		ET 200SP HA, IM155–6 PN															SIEMENS				
A0	6DL1193–6AR00–0AA0		ET 200SP HA, ADATTATORE DI BUS BA 2XRJ45															SIEMENS				
A0	6DL1193–6BH00–0SM0		MODULO SUPPORTO IM SINGLE															SIEMENS				
A1	6ES7137–6AA01–0BA0		ET 200SP, CM PTP, VPE 1															SIEMENS				
A1	6ES7193–6BP00–0DA0		BaseUnit BU15–P16+A0+2D, BU type A0, push–in terminals,															SIEMENS				
A10	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A10	6DL1132–6EB00–0HX1		ET 200SP HA, EX–DQ 2x23,1VDC/20MA															SIEMENS				
A11	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A11	6DL1132–6EB00–0HX1		ET 200SP HA, EX–DQ 2x23,1VDC/20MA															SIEMENS				
A2	6DL1193–6BP00–0DW0		ET 200SP HA, Ex–BU TYPE W0															SIEMENS				
A2	6DL1134–6TB00–0HX1		ET 200SP HA, EX–AI 2xI 2–WIRE HART															SIEMENS				
A3	6DL1193–6BP00–0DW0		ET 200SP HA, Ex–BU TYPE W0															SIEMENS				
A3	6DL1134–6TB00–0HX1		ET 200SP HA, EX–AI 2xI 2–WIRE HART															SIEMENS				
A4	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A4	6DL1134–6JD00–0HX1		ET 200SP HA, EX–AI 4xTC/2xRTD 2–/3–/4–W															SIEMENS				
A5	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A5	6DL1134–6JD00–0HX1		ET 200SP HA, EX–AI 4xTC/2xRTD 2–/3–/4–W															SIEMENS				
A6	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A6	6DL1135–6TB00–0HX1		ET 200SP HA, EX–AQ 2xI HART															SIEMENS				
A7	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A7	6DL1131–6TD00–0HX1		ET 200SP HA, EX–DI 4xNAMUR															SIEMENS				
A8	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A8	6DL1131–6TD00–0HX1		ET 200SP HA, EX–DI 4xNAMUR															SIEMENS				
A9	6DL1193–6BP00–0BX1		ET 200SP HA, Ex–BU TYPE X1															SIEMENS				
A9	6DL1132–6EB00–0HX1		ET 200SP HA, EX–DQ 2x23,1VDC/20MA															SIEMENS				
ESD	M–0445		RED EMERGENCY PUSH–BUTTON , TURN TO RELEASE, 1 N.O. + 1 N.C. CONTACTS															CORTEM				
F00	952110		DG M TT 2P 275 – SPD															DEHN				
F1	2302038		Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V															ITALWEBER				
F1	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE				
F2	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE				
F3	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE				
F4	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE				
F5	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE				
FG1	2301138		Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V															ITALWEBER				
FRK0	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE				
FXC	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE				
G1	2903149		TRIO–PS–2G/1AC/24DC/10 – Power supply unit															PHOENIX CONTACT				
G2	6DL1133–6PX00–0HWO		ET 200SP HA, Ex–PM E POWER MODULE															SIEMENS				
HS–S–221AB	M–0553		EXTERNAL HANDLE															CORTEM				
HS–S–221AB	U2–10E/R		”SONTHEIMER SEL.SWITCH ””1–0–2””,2P,10A, WITH 3 FIXED POSITIONS”															CORTEM				
HS–S221AC	M–0553		EXTERNAL HANDLE															CORTEM				
HS–S221AC	WS2–10E/R		”SONTHEIMER SEL.SWITCH ””1–2””,2P,10A, WITH 2 FIXED POSITIONS (SEE DETAIL)”															CORTEM				
HS–S221BC	M–0553		EXTERNAL HANDLE															CORTEM				
HS–S221BC	WS2–10E/R		”SONTHEIMER SEL.SWITCH ””1–2””,2P,10A, WITH 2 FIXED POSITIONS (SEE DETAIL)”															CORTEM				
HS–S–221S0	M–0429/R		RED PUSH–BUTTON 1 N.O. + 1 N.C.															CORTEM				
HS–S–221SA	M–0429/V		GREEN PUSH–BUTTON 1 N.O. + 1 N.C.															CORTEM				
P–PN–S221B.1	EJB–7		ENCLOSURE 1000x700x510 mm L x H x D															CORTEM				
P–PN–S221B.1	BFE–7		INTERNAL FRAME															CORTEM				
Q0	M–0553L		EXTERNAL HANDLE, PADLOCKABLE															CORTEM				
Q0	A3–10E/R		”SONTHEIMER NOT AUTOMATIC SWITCH ””ON–OFF””,3P,10A, PADLOCKABLE IN ””OFF”” POSITION”															CORTEM				
XL–P–221AA	M–0457/3R24		SIGNALLING LED RED COLOUR 24VDC															CORTEM				
XL–P–221AR	M–0457/3I24		SIGNALLING LED WHITE COLOUR 230V															CORTEM				
XL–P–221AU	M–0457/3V24		SIGNALLING LED GREEN COLOUR 24VDC															CORTEM				
XL–P–221BA	M–0457/3R24		SIGNALLING LED RED COLOUR 24VDC															CORTEM				
XL–P–221BR	M–0457/3I24		SIGNALLING LED WHITE COLOUR 230V															CORTEM				
XL–P–221BU	M–0457/3V24		SIGNALLING LED GREEN COLOUR 24VDC															CORTEM				
General Contractor : Manufacture : 			Project : P–PN–S221B.1 LNG LOADING BAY S–221B			Customer :  Contract : IE24090 S/N : IE24090_SE03						Object :										
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